Sampling Theory and Testing Hypothesis

1. Collection of individuals – **Population (or) Universe**

2. Population may be **finite (or) infinite**.

3. Finite subset of individuals in a population – **Sample**

4. Number of individuals in a sample – **Sample size**

5. Statistical measure [mean and variance] calculated on the basis of Population values of x - **Parameters**

6. Statistical measure computed on the basis of sample observations $(x_1, x_2, ..., x_n)$ - **Statistics.**

7. A procedure for deciding whether to accept (or) to reject a null hypothesis {and hence to reject (or) to accept the alternative hypothesis} – **Testing of Hypothesis.**

8. Assuming that there is no significant difference between sample statistic and corresponding population parameter (or) between two sample statistics – **Null Hypothesis** H_o.

9. Hypothesis different from null hypothesis – **Alternative Hypothesis,** H_1

10. A procedure for testing the hypothesis whether the difference between population parameter θ_0 and corresponding sample statistic θ is significant (or) not – **Test of Significance.**

11. **Errors in Sampling:**

    i) Type I Error (Producer's Risk): Reject H_o when it is true.

    ii) Type II Error (Consumer's Risk): Accept H_o when it is wrong

Actual	Decision	
	Accept H_o	Reject H_o
H_o is true	Correct decision (no error) Probability = $(1-\alpha)$	Wrong Type I Error Probability α
H_o is false	Wrong Type II Error Probability β	Correct decision (no error) Probability = $(1-\beta)$

12. **One Tailed and Two Tailed Tests:**

- A test of any statistical hypothesis where the alternative hypothesis is one – tailed (right tailed or left tailed) is called a *One – Tailed test.*

  For Example: For testing the mean of population in a single tailed, we assume that the
  null hypothesis :H_0 : $\mu = \mu_0$ against the
  alternative hypothesis : H_1 : $\mu > \mu_0$ (Right tailed)
  H_1 : $\mu < \mu_0$ (Left tailed) is called One – tailed test.

- In a test of statistical hypothesis where the alternative hypothesis is 2- tailed, we assume that the
  null hypothesis : H_0 : $\mu = \mu_0$ against the
  alternative hypothesis :H_1 : $\mu \neq \mu_0$ is called *2- tailed test.*

- Suppose if we want to test if the bulbs produced by new process (μ_2) have *higher average life* than those produced by standard process (μ_1) then we've H_0 : $\mu_1 = \mu_2$ and H_1 : $\mu_1 < \mu_2$ (left tail test)

- If we want to test whether the product of new process (μ_2) is *inferior* to that of standard process (μ_1) then H_0 : $\mu_1 = \mu_2$ and H_1 : $\mu_1 > \mu_2$ (right tail test)

13**. Critical values of Z_α :**

Critical values (z_α)	Level of Significance (α)			
	1%	2%	5%	10%
Two tailed test	$\lvert z_\alpha \rvert = 2.58$	$\lvert z_\alpha \rvert = 2.33$	$\lvert z_\alpha \rvert = 1.96$	$\lvert z_\alpha \rvert = 1.645$
Right tailed test	$z_\alpha = 2.33$	$z_\alpha = 2.054$	$z_\alpha = 1.645$	$z_\alpha = 1.28$
Left tailed test	$z_\alpha = -2.33$	$z_\alpha = -2.054$	$z_\alpha = -1.645$	$z_\alpha = -1.28$

14. **Procedure for Testing of Hypothesis:**

a. Set up Null hypothesis $H_o : \mu = \mu_o$
b. Set up Alternative hypothesis $H_1 : \mu \neq \mu_o$
c. Test Statistic (Calculated Value) $Z = \dfrac{t - E(t)}{S.E(t)}$
d. Level of Significance (Tabulated value) Z_α : 1% (or) 5%
e. Conclusion: a) If $|Z| < Z_\alpha$ then accept H_o
   b) If $|Z| > Z_\alpha$ then reject H_o

15. A region corresponding to a statistic t in the sample space S which lead to the rejection of H_o - **Critical Region (or) Rejection Region**

16. The region which lead to the acceptance of H_o - **Acceptance Region**

17. **Test of Significance of Small Samples:**

When sample size is less than 30 $(n < 30)$ then the sample is called as Small sample.

1. Student's – t test for single mean:

Null hypothesis : $H_o : \mu = \mu_o$

Alternative hypothesis: $H_1 : \mu \neq \mu_o$

Test statistic: $t = \dfrac{\bar{x} - \mu}{S^2 / \sqrt{n}}$ (S.D is not given directly)

Where $\bar{x} = \dfrac{\sum x_i}{n}$ = sample mean; μ = population mean

$S^2 = \dfrac{\sum (x_i - \bar{x})^2}{n-1}$; n = sample size

ii) Test statistic : $t = \dfrac{\bar{x} - \mu}{S.D / \sqrt{n-1}}$ (S.D is given directly)

Degree of freedom(d.f) : $n-1$

Confidence (or) Fiducial limits for μ

1. The 95% confidence limit of the mean of the population is given by $\bar{x} \pm t_{0.05} \dfrac{S}{\sqrt{n}}$.
2. The 99% confidence limit of the mean of the population is given by $\bar{x} \pm t_{0.01} \dfrac{S}{\sqrt{n}}$

Problems:

1. The mean monthly sale of a particular household item in department stores was observed as 133.3 units per store. After an advertising campaign, data on sales of this item from 25 stores were collected and it is found that the mean sales has increased to 141.5 units with a standard deviation of 15.2

units. Was the advertising campaign successful? Assume that level of significance is 5%.

Solution: Given: Sample size $n = 25$, Sample mean $\bar{x} = 141.5$
S.D (S) =15.2, Population mean $\mu = 133.3$
Degree of freedom = $n - 1$ = 24

Null hypothesis : $H_o : \mu = 133.3$ Not Successful

Alternative hypothesis: $H_1 : \mu > 133.3$

Test statistic (C.V) : $t = \dfrac{\bar{x} - \mu}{S.D/\sqrt{n-1}} = \dfrac{141.5 - 133.3}{15.2/\sqrt{24}} = \dfrac{8.20}{3.10} = 2.65$

Tabulated Value (T.V) : $t_{0.05}$ with 24 d.f is 2.06

Conclusion : C.V >T.V
we reject the null hypothesis H_o.
The advertising campaign was successful.

2. The mean life time of a sample 25 fluorescent light bulbs produced by a company is computed to be 1570 hours with a S.D of 120 hours. The company claims that the average life of the bulbs produced by the company is 1600 hours. Use the level of significance of 5%. Is the claim acceptable?

Solution: Given: Sample size $n = 25$, Sample mean $\bar{x} = 1570$
S.D (S) = 120 , Population mean $\mu = 1600$
Degree of freedom = $n - 1$ = 24

Null hypothesis : $H_o : \mu = 1600\ hrs$

Alternative hypothesis: $H_1 : \mu \neq 1600\ hrs$

Test statistic (C.V) : $t = \dfrac{\bar{x} - \mu}{S.D/\sqrt{n-1}} = \dfrac{1570 - 1600}{120/\sqrt{24}} = \dfrac{-30}{24.49}$ = -1.22

$|t| = 1.22$

Tabulated Value (T.V) : $t_{0.05}$ with 24 d.f is 2.06

Conclusion : C.V < T.V

we accept the null hypothesis H_o.

The claim is acceptable

3. A machinist is making engine parts with axle diameters of 0.700 inch. A random sample of 10 parts shows a mean diameter of 0.742 inch with a S.D of 0.040 inch. Compute the statistic you would use to test whether the work is meeting the specification.

Solution: Given: Sample size $n=10$, Sample mean $\bar{x}=0.742$

S.D (S) =0.040, Population mean $\mu=0.700$

Degree of freedom = $n-1$ = 9

Null hypothesis : $H_o : \mu = 0.700$

Alternative hypothesis: $H_1 : \mu \neq 0.700$

Test statistic (C.V) : $t = \dfrac{\bar{x}-\mu}{S.D/\sqrt{n-1}} = \dfrac{0.742-0.700}{0.040/\sqrt{9}} = 3.15$

$|t| = 3.15$

Tabulated Value (T.V) : $t_{0.05}$ with 9 d.f is 2.26

Conclusion : C.V > T.V

we reject the null hypothesis H_o.

4. A random sample of size 16 valves from a normal population showed a mean of 53 and a sum of squares of deviation from the mean equals to 150. Can this sample be regarded as taken from the population having 56 as mean? Obtain 95% confidence limits of the mean of the population.

Solution: Given: Sample size $n=16$, Sample mean $\bar{x}=53$

$\sum(x-\bar{x})^2 = 150$, Population mean $\mu = 56$

Therefore, $S^2 = \dfrac{\sum(x-\bar{x})^2}{n-1} = \dfrac{150}{15} = 10$

$\Rightarrow S = \sqrt{10}$

Degree of freedom = $n-1$ = 15

Null hypothesis : $H_o : \mu = 56$

Alternative hypothesis: $H_1 : \mu \neq 56$

Test statistic (C.V) : $t = \dfrac{\bar{x} - \mu}{S/\sqrt{n}} = \dfrac{53-56}{\sqrt{10}/\sqrt{16}}$ = -3.79

$|t| = 3.79$

Tabulated Value (T.V) : $t_{0.05}$ with 15 d.f is 2.13

Conclusion : C.V > T.V

we reject the null hypothesis H_o.

The 95% confidence limit of the mean of the population is given by

$$\bar{x} \pm t_{0.05} \frac{S}{\sqrt{n}} = 53 \pm 2.13 \times 0.79$$

$$= 53 \pm 1.6827$$

Hence 95% confidence limit is [51.31, 54.68]

5. A random sample of 10 boys had the following I.Q's 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of a population mean I.Q. of 100? Find a reasonable range in which most of the mean I.Q. values of samples of 10 boys lie.

Solution:

x	$x - \bar{x}$	$(x - \bar{x})^2$
70	-27.2	739.84
120	22.8	519.84
110	12.8	163.84
101	3.8	14.44
88	-9.2	84.64
83	-14.2	201.64
95	-2.2	4.84
98	0.8	0.64
107	9.8	96.04
100	2.8	7.84
972		1833.60

Mean $\bar{x} = \dfrac{\sum x}{n} = \dfrac{972}{10} = 97.2$

We know that, $S^2 = \dfrac{\sum (x_i - \bar{x})^2}{n-1} = \dfrac{1833.60}{9}$ = 203.73

Standard deviation, $S = \sqrt{203.73}$ = 14.27

Null hypothesis : $H_o : \mu = 100$

Alternative hypothesis: $H_1 : \mu \neq 100$

Test statistic (C.V) : $t = \dfrac{\bar{x} - \mu}{S/\sqrt{n}} = \dfrac{97.2 - 100}{14.27/\sqrt{10}}$ = -0.62

$|t| = 0.62$

Tabulated Value (T.V) : $t_{0.05}$ with 9 d.f is 2.26

Conclusion : C.V < T.V

we accept the null hypothesis H_o.

The 95% confidence limit of the mean of the population is given by

$$\bar{x} \pm t_{0.05} \frac{S}{\sqrt{n}} = 97.2 \pm 2.26 \times 4.514$$

$$= 97.2 \pm 10.20$$

Hence 95% confidence limit is [87.00, 107.40]

2. Students t – test for difference of means:

To test the significant difference between two means $\bar{x}_1$ and $\bar{x}_2$ of sample of sizes n_1 and n_2.

Null Hypothesis: $H_o : \mu_1 = \mu_2$

Alternative hypothesis: $H_1 : \mu_1 \neq \mu_2$

Test statistic: $t = \dfrac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}}$

Where $s^2 = \dfrac{\sum(x_1 - \bar{x}_1)^2 + \sum(x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}$ [S.D is not given directly]

$s^2 = \dfrac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$ [S.D is given directly]

s_1, s_2 - sample standard deviation.

Degrees of freedom $d.f. = n_1 + n_2 - 2$

1. The average number of articles produced by two machines per day are 200 and 250 with standard deviations 20 and 25 respectively on the basis of records of 25 days production. Can you regard both the machines equally efficient at 1% level of significance?

Solution: Given: $n_1 = 25$, $\bar{x}_1 = 200$, $s_1 = 20$

$n_2 = 25$, $\bar{x}_2 = 250$, $s_2 = 25$

Now, $s^2 = \dfrac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} = \dfrac{25(20)^2 + 25(25)^2}{48} = 533.85$

$\Rightarrow \quad s = 23.10$

Null Hypothesis: $H_o : \mu_1 = \mu_2$

Alternative hypothesis: $H_1 : \mu_1 \neq \mu_2$

Test statistic (C.V) : $t = \dfrac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}} = \dfrac{200 - 250}{23.10\sqrt{\dfrac{1}{25} + \dfrac{1}{25}}} = -7.65$

$|t| = 7.65$

Tabulated value (T.V): $t_{0.01}$ for 48d.f. is 2.58

Conclusion : C.V > T.V

We reject the null hypothesis H_o.

2. The means of two random samples of size 9 and 7 are 196.42 and 198.82 respectively. The sum of the squares of the deviation from the mean are 26.94 and 18.73 respectively can the sample be considered to have been drawn from the same normal population.

Solution: Given: $n_1 = 9$, $\bar{x}_1 = 196.42$, $\sum(x_1 - \bar{x}_1)^2 = 26.94$

$n_2 = 7$, $\bar{x}_2 = 198.82$, $\sum(x_2 - \bar{x}_2)^2 = 18.73$

Now, $s^2 = \dfrac{\sum(x_1 - \bar{x}_1)^2 + \sum(x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2} = \dfrac{26.94 + 18.73}{9 + 7 - 2} = 3.26$

$\Rightarrow \quad s = 1.81$

Null Hypothesis: $H_o : \mu_1 = \mu_2$

Alternative hypothesis: $H_1 : \mu_1 \neq \mu_2$

Test statistic (C.V) : $t = \dfrac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}} = \dfrac{196.42 - 198.82}{1.81\sqrt{\dfrac{1}{9} + \dfrac{1}{7}}} = -2.63$

$|t| = 2.63$

Tabulated value (T.V): $t_{0.05}$ for 14 d.f. is 2.14

Conclusion : C.V > T.V

We reject the null hypothesis H_o.

3. The nicotine content in milligrams of two samples of tobacco were found to be as follows:

Sample A	24	27	26	21	25	-
Sample B	27	30	28	31	22	36

Can it be said that two samples come from normal populations having the same mean.

Solution:

Sample A (x)	Sample B (y)	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(y-\bar{y})^2$
24	27	-0.6	0.36	-2	4
27	30	2.4	5.76	1	1
26	28	1.4	1.96	-1	1
21	31	-3.6	12.96	2	4
25	22	0.4	0.16	-7	49
	36			7	49
123	**174**	**0**		**0**	**108**

Mean $\bar{x} = \frac{123}{5} = 24.6$, $\bar{y} = \frac{174}{6} = 29$

$\sum (x-\bar{x})^2 = 21.2$, $\sum (y-\bar{y})^2 = 108$

Now, $s^2 = \frac{\sum (x-\bar{x})^2 + \sum (y-\bar{y})^2}{n_1 + n_2 - 2} = \frac{21.2+108}{5+6-2} = 14.35$

$\Rightarrow s = 3.78$

Null Hypothesis : $H_o : \mu_1 = \mu_2$

Alternative hypothesis: $H_1 : \mu_1 \neq \mu_2$

Test statistic (C.V) : $t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{24.6 - 29}{3.78\sqrt{\frac{1}{5} + \frac{1}{6}}} = -1.92$

$|t| = 1.92$

Tabulated value (T.V): $t_{0.05}$ for 9 d.f. is 2.26

Conclusion : C.V < T.V

We accept the null hypothesis H_o.

3. F- test: (To test Equality of variance)

Null Hypothesis : $H_0 : \sigma_1^2 = \sigma_2^2$ {(ie) Population variance are same}

Alternative hypothesis : $H_1 : \sigma_1^2 \neq \sigma_2^2$

Test Statistic: $F = \dfrac{S_1^2}{S_2^2}$, $S_1^2 > S_2^2$

where $S_1^2 = \dfrac{\sum (x-\bar{x})^2}{n_1 - 1}$, $S_2^2 = \dfrac{\sum (y-\bar{y})^2}{n_2 - 1}$ {S.D is not given directly}

$S_1^2 = \dfrac{n_1 s_1^2}{n_1 - 1}$, $S_2^2 = \dfrac{n_2 s_2^2}{n_2 - 1}$ { S.D is given directly}

Degree of freedom : $\gamma_1 = n_1 - 1$ and $\gamma_2 = n_2 - 1$

Note:. $F = \dfrac{Greater\ variance}{Smaller\ variance}$

1. In one sample of 10 observations from a normal population, the sum of the squares of the deviations of the sample values from the sample mean is 102.4 and in another sample of 12 observations from another normal population, the sum of the squares of the deviations of the sample values from the sample mean is 120.5. Examine whether the two normal populations have the same variances.

   Solution:

   Given: $n_1 = 10$, $\sum (x_1 - \bar{x}_1)^2 = 102.4$, $s_1^2 = \dfrac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1} = \dfrac{102.4}{9} = 11.37$

   $n_2 = 12$, $\sum (x_2 - \bar{x}_2)^2 = 120.5.4$, $s_2^2 = \dfrac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1} = \dfrac{120.5}{11} = 10.95$

Null Hypothesis : $H_0 : \sigma_1^2 = \sigma_2^2$

Alternative hypothesis : $H_1 : \sigma_1^2 \neq \sigma_2^2$

Test Statistic (C.V) : $F = \dfrac{s_1^2}{s_2^2} = \dfrac{11.37}{10.95} = 1.038$

Tabulated Value (T.V) : $F_{0.05}$ (9,11) d.f is 2.91

Conclusion : C.V < T.V

We accept the null hypothesis H_o.

2. Is known that the mean diameters of rivets produces by two firms A and B are practically the same but the standard deviations may differ. For 22 rivets produced by firm A, the standard deviation is 2.9 mm, where 16 rivets manufactured by firm B, the standard deviation is 3.8 mm. Compute the

statistic you would use to test whether the products of firm A have the same variability as those of firm B and test its significance.

Solution: Given : $n_1 = 22$, $s_1 = 2.9\ mm$

$n_2 = 16$, $s_2 = 3.8\ mm$

Here the S.D's of the samples s_1 and s_2 are given.

Therefore, the population variance S_1^2 and S_2^2 are obtained by using the relations. $n_1 s_1^2 = (n_1 - 1) S_1^2$ and $n_2 s_2^2 = (n_2 - 1) S_2^2$

$$\Rightarrow S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} \qquad \Rightarrow S_2^2 = \frac{n_2 s_2^2}{n_2 - 1}$$

$$= \frac{22 \times (2.9)^2}{21} \qquad = \frac{16 \times (3.8)^2}{15}$$

$$= 8.8105 \qquad = 15.403$$

Null Hypothesis : $H_0 : \sigma_1^2 = \sigma_2^2$

Alternative hypothesis : $H_1 : \sigma_1^2 \neq \sigma_2^2$

Test Statistic (C.V) : $F = \frac{s_1^2}{s_2^2} = \frac{15.403}{8.8105} = 1.75$

Tabulated Value (T.V) : $F_{0.05}$ (15,21) d.f is 2.18

Conclusion : C.V < T.V

We accept the null hypothesis H_o.

3. The nicotine contents in milligrams in two samples of tobacco were found to be as follows:

Sample A	24	27	26	21	25	-
Sample B	27	30	28	31	22	36

Can it be said that two samples come from same normal population?

Solution: Hint: To test whether the two samples come from the same normal population.

That is to test i) the equality of variances

ii) equality of means

Chi – Square test (The square of a standard normal variate)

A test for testing the significance of difference between experimental values and the theoretical values obtained under some theory or hypothesis.

This test is known as ψ^2 - test of goodness of fit.

Test Statistic:

$$\psi^2 = \sum \frac{(O-E)^2}{E}$$ where O - Observed frequency

E – Expected frequency

Note:

1. It is used to test the difference between observed and expected frequencies are significant.
2. If the data is given in a series of 'n' number then degrees of freedom = n-1
3. In case of a .Binomial distribution d.f = n-1
   b. Poisson distribution d.f. = n-2
   c. Normal distribution d.f. = n-3

1. The following table gives the number of aircraft accidents that occurred during the various days of the week. Test whether the accidents are uniformly distributed over the week.

Days	Mon	Tue	Wed	Thur	Fri	Sat
No. of accidents	14	18	12	11	15	14

Solution:

Null hypothesis H_o : The accidents are uniformly distributed over the week.

Expected frequencies of the accidents on each of the days = $\frac{84}{6} = 14$

Observed Frequency (O)	Expected Frequency (E)	$(O-E)^2$	$\frac{(O-E)^2}{E}$
14	14	0	0
18	14	16	1.143
12	14	4	0.286
11	14	9	0.643
15	14	1	0.071
14	14	0	0
			2.143

Test Statistic (C.V) : $\psi^2 = \sum \frac{(O-E)^2}{E}$ = 2.143

Degrees of freedom (d.f) : $n-1$ = 6-1 = 5

Tabulated Value (T.V) : ψ^2 at 5% level for 5 d.f. = 11.07

Conclusion : C.V < T.V

We accept the null hypothesis H_o.

Therefore, the accidents are uniformly distributed over the week.

2. A sample analysis of examination results of 500 students was made. It was found that 220 students had failed, 170 had secured a third class, 90 were placed in second class and 20 got a first class. Do these figures commensurate with the general examination result which is in the ratio of 4: 3: 2: 1 for the various categories respectively.

Solution:
Null hypothesis H_o : The observed results commensurate with the general examination results.
Expected frequencies are in the ration 4: 3: 2: 1
Total frequency = 500
Expected frequency = Dividing the total frequency in the ratio 4: 3: 2: 1

$$= \frac{4}{10} \times 500 = 200,\ \frac{3}{10} \times 500 = 150,\ \frac{2}{10} \times 500 = 100$$

$$\frac{1}{10} \times 500 = 50$$

Class	Observed Frequency (O)	Expected Frequency (E)	$(O-E)^2$	$\frac{(O-E)^2}{E}$
Failed	220	200	400	2.000
Third	170	150	400	2.667
Second	90	100	100	1.000
First	20	50	900	18.000
				23.667

Test statistic (C.V) : $\psi^2 = \sum \frac{(O-E)^2}{E}$ = 23.667
Degrees of freedom (d.f) : $n-1$ = 4-1 = 3
Tabulated Value (T.V) : ψ^2 at 5% level for 3 d.f. = 7.81
Conclusion : C.V > T.V
We reject the null hypothesis H_o.
Therefore, the observed results are not commensurate with the general examination results.

Chi- Square Test for Independence of Attributes:

Attribute means a quality of characteristics.(Eg: dancing, honesty etc.,)

An attribute may be marked by its presence (position) or absence in a number of a given population.

Let us consider 2 attribute A & B. A is divided into 2 classes and B is divided into 2 classes. The various cell frequencies can be expressed in the following table known as 2 x 2 contingency table.

A	a	b
B	c	d

a	b	a + b
c	d	c + d
a + c	b + d	N

The Expected frequencies are given by

E(a) = $\frac{(a+c)(a+b)}{N}$	$E(b) = \frac{(b+d)(a+b)}{N}$	a + b
$E(c) = \frac{(a+c)(c+d)}{N}$	$E(d) = \frac{(b+d)(c+d)}{N}$	c + d
a + c	b + d	N (Total frequencies)

Note:

1. Degree of freedom : $(r-1)(s-1)$ where r – number of rows, s – number of columns
2. Null hypothesis H_0 : Attributes are independent.
3. Testing that if 2 attributes A & B under consideration are independent (or) not.

1. The following table gives a classification of a sample of 100 plants of their flower colour and flatness of leaf.

	Flat leaves	Curved leaves	Total
White flower	40	20	60
Red flower	10	30	40
Total	50	50	100

Test whether the flower colours is independent of the flatness of leaf.

Solution:

Null hypothesis H_0 : They are independent.

Expected frequencies are

$\frac{50 \times 60}{100} = 30$	$\frac{50 \times 60}{100} = 30$	60
$\frac{50 \times 40}{100} = 20$	$\frac{50 \times 40}{100} = 20$	40

50	50	100

Now,

Observed Frequency (O)	Expected Frequency (E)	$(O-E)^2$	$\frac{(O-E)^2}{E}$
40	30	100	3.333
20	30	100	3.333
10	20	100	5.000
30	20	100	5.000
			16.666

Test statistic (C.V) : $\psi^2 = \sum \frac{(O-E)^2}{E}$ = 16.666

Degrees of freedom (d.f) : = $(r-1)(s-1)$ = (2-1)(2-1) = 1

Tabulated Value (T.V) : ψ^2 at 5% level for 1 d.f. = 3.84

Conclusion : C.V > T.V

We reject the null hypothesis H_o.

Large Samples

If the sample size n > 30 then the sample is called Large sample.

1. **Test of Significance for Single Proportion:**

To test the significant difference between the sample proportion p and population proportion P.

Test Statistic:

$$Z = \frac{p-P}{\sqrt{\frac{PQ}{n}}}$$ where n – sample size

Note:

1. Limits for population proportion P are given by $p \pm 3\sqrt{pq/n}$ where $q = 1 - p$

2. 98% confidence limits for population proportion are $p \pm 2.33\sqrt{pq/n}$

1. In a city, a sample of 1000 people were taken and out of them 540 are vegetarians and the rest are non-vegetarians. Can we say that both habits of eating (vegetarian or non-vegetarian) are equally popular in the city at 1% level of significance.

Solution:

Given: $n = 1000$, p = sample proportion of vegetarians = $\frac{540}{1000} = 0.54$,

P = population proportion of vegetarians = $\frac{1}{2}$ =0.5

Null hypothesis H_0 : Both habits are equally popular. ($H_o : P = 0.5$)

Alternative hypothesis : $H_1 : P \neq 0.5$

Test Statistic (C.V) : $Z = \frac{p-P}{\sqrt{\frac{PQ}{n}}} = \frac{0.54-0.5}{\sqrt{\frac{0.5\times 0.5}{1000}}}$ = 2.532

Tabulated value (T.V) : Z at 1% level of significance is 2.58

Conclusion : C.V < T.V

We accept null hypothesis H_o.

2. A die was thrown 9000 times and of these 3220 yielded a 3 or 4. Is this consistent with the hypothesis that the die was unbiased?

Solution: Given: $n = 9000$,

p = Proportion of successes of getting 3 or 4 in 9000 throws = $\frac{3220}{9000} = 0.3578$,

P = population proportion of successes = P(getting a 3 or 4) = $\frac{1}{6}+\frac{1}{6}=\frac{2}{6}$ =0.33

Null hypothesis H_0 : The die is unbiased. ($H_o : P = \frac{1}{3}$)

Alternative hypothesis : $H_1 : P \neq \frac{1}{3}$

Test Statistic (C.V) : $Z = \frac{p-P}{\sqrt{\frac{PQ}{n}}} = \frac{0.3578-0.3333}{\sqrt{\frac{0.3333\times 0.6667}{9000}}}$ = 4.94

Tabulated value (T.V) : Z at 1% level of significance is 2.58

Conclusion : C.V > T.V

We reject null hypothesis H_o.

2. Test of Significance of Difference of Proportions:

To test the significant difference between the sample proportions p_1 and p_2.

Null hypothesis is : $H_0 : p_1 = p_2$

Alternative hypothesis is : $H_1 : p_1 \neq p_2$

a) To test the significant difference between the sample proportions p_1 & p_2

Test Statistic : $Z = \dfrac{p_1 - p_2}{\sqrt{pq\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}}$ where $p = \dfrac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$

and $q = 1 - p$

b) To test the significant difference between p_1 & p

Test Statistic : $Z = \dfrac{p_1 - p}{\sqrt{\dfrac{n_2 pq}{n_1(n_1 + n_2)}}}$ where $p = \dfrac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$

c) If the sample proportions are not known then $Z = \dfrac{P_1 - P_2}{\sqrt{\dfrac{P_1 Q_1}{n_1} + \dfrac{P_2 Q_2}{n_2}}}$

1. Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of the proposal. Test the hypothesis that proportions of men and women in favour of the proposal are same, as 5% level.

Solution:

Given: sample sizes $n_1 = 400, \ n_2 = 600.$

Proportion of men = $p_1 = \dfrac{200}{400} = 0.5$

Proportion of women = $p_2 = \dfrac{325}{600} = 0.541$

Null hypothesis : $H_0 : p_1 = p_2$

Alternative hypothesis : $H_1 : p_1 \neq p_2$

Test Statistic : $Z = \dfrac{p_1 - p_2}{\sqrt{pq\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}} = \dfrac{0.5 - 0.541}{\sqrt{0.525 \times 0.425\left(\dfrac{1}{400} + \dfrac{1}{600}\right)}}$

$$= \frac{-0.041}{0.032} = -1.34$$

$|z| = 1.34$

{where $p = \dfrac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \dfrac{400 \times \frac{200}{400} + 600 \times \frac{325}{600}}{400 + 600} = \dfrac{525}{1000} = 0.525$ }

Tabulated value (T.V) : Z at 5% level of significance is 1.96
Conclusion : C.V < T.V
We accept null hypothesis H_o.

2. In a random sample of 400 student of the university teaching department, it was found that 300 students failed in the examination. In another sample of 500 students of the affiliated colleges the number of failures in the same examination was found to be 300. Find out whether the proportion of failures in the university teaching departments significantly greater than the proportion of failures in the university teaching departments and affiliated colleges taken together.

Solution:

Given: $n_1 = 400,\ n_2 = 500,\ p_1 = \dfrac{300}{400} = 0.75,\ p_2 = \dfrac{300}{500} = 0.6$

$$p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{400 \times 0.75 + 500 \times 0.6}{400 + 500} = 0.667\ ,\ q = 0.333$$

Null hypothesis H_o : Assume that there is no significant difference between p_1 & p

Test Statistic : $Z = \dfrac{p_1 - p}{\sqrt{\dfrac{n_2 pq}{n_1(n_1 + n_2)}}} = \dfrac{0.75 - 0.667}{\sqrt{\dfrac{500 \times 0.667 \times 0.333}{400(400 + 500)}}} = 4.74$

Tabulated value : z at 5% level of significance is 1.96
Conclusion : C.V > T.V
We reject the null hypothesis H_o.

Therefore, the proportion of failures in the affiliated colleges is greater than the proportion of failures in university departments and affiliated colleges taken together.

3. In two large populations, there are 30% and 25% respectively of fair haired people. Is this difference likely to be hidden in samples of 1200 and 900 respectively from the two populations.

Solution:

Given: $n_1 = 1200,\ n_2 = 900$,

P_1 = Proportion of fair hairedd people in the first population = $\dfrac{30}{100} = 0.3$

P_2 = Proportion of fair hairedd people in the second population = $\frac{25}{100} = 0.25$

Null hypothesis H_o : Assume that the sample proportions are equal.

$H_0 : p_1 = p_2$

Alternative hypothesis : $H_1 : p_1 \neq p_2$

Test Statistic : $Z = \frac{P_1 - P_2}{\sqrt{\frac{P_1Q_1}{n_1} + \frac{P_2Q_2}{n_2}}} = \frac{0.3 - 0.25}{\sqrt{\frac{0.3 \times 0.7}{1200} + \frac{0.25 \times 0.75}{900}}} = 2.55$

Tabulated value : z at 5% level of significance is 1.96

Conclusion : C.V > T.V

We reject the null hypothesis H_o.

Therefore, the proportion are not equal.

3. Test of Significance for Single Mean:

To test whether the given sample of size n has been drawn from a population with mean μ .

Null Hypothesis: There is no difference between sample mean $\bar{x}$ and population mean μ

Test Statistic: $Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$ where σ - S.D of the population

Note:

1. If the population S.D is not known then the test statistic

$Z = \frac{\bar{x} - \mu}{s / \sqrt{n}}$ where s – sample S.D

2. 95% confidence limits for μ are $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$

3. 98% confidence limits for μ are $\bar{x} \pm 2.33 \frac{\sigma}{\sqrt{n}}$

4. 99% confidence limits for μ are $\bar{x} \pm 2.58 \frac{\sigma}{\sqrt{n}}$

1. A sample of 900 members has a mean of 3.4 cms and a S.D. 2.61 cms. Is the sample from a large population of mean 3.25cm and a S.D. 2.61 cms. If the population is normal and its mean is unknown find the 95% confidence limits of true mean.

Solution: Given : $n = 900$, $\mu = 3.25$, $\bar{x} = 3.4cm$, $\sigma = 2.61$, $s = 2.61$

Null hypothesis : $H_o : \mu = 3.25$

Alternative hypothesis: $H_1 : \mu \neq 3.25$

Test statistic (C.V) : $z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \dfrac{3.4 - 3.25}{2.61/\sqrt{900}} = 1.724$

Z = 1.724

Tabulated Value (T.V) : Z at 5% level of significance is 1.96

Conclusion : C.V < T.V

we accept the null hypothesis H_o.

Therefore, the samples has been drawn from the population mean $\mu = 3.25$

The 95% confidence limit of the mean of the population is given by $\bar{x} \pm 1.96 \dfrac{\sigma}{\sqrt{n}}$

$= 3.4 \pm 1.96 \times \dfrac{2.61}{\sqrt{900}} = 3.4 \pm 0.1705$

Hence 95% confidence limit is [3.2295, 3.57]

4. Test of Significance for Difference of Mean:

1. To test the significant difference between $\bar{x}_1$ and $\bar{x}_2$

Test Statistic: $Z = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}$

2. If the samples have been drawn from the same population then $\sigma_1^2 = \sigma_2^2 = \sigma^2$

Test Statistic: $Z = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{\sigma^2}{n_1} + \dfrac{\sigma^2}{n_2}}}$

3. If σ is not known, we can use a estimate of σ^2 given by $\sigma^2 = \dfrac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}$

1. The means of 2 large samples 1000 and 2000 members are 67.5 inches and 68.0 inches respectively. Can the samples be regarded as drawn from the same population of S.D. 2.5 inches.

Solution : Given: Sample sizes $n_1 = 1000$, $n_2 = 2000$.

Sample mean $\bar{x}_1 = 67.5\ inches$, $\bar{x}_2 = 68\ inches$

Population S.D $\sigma = 2.5\ inches$

Null Hypothesis : $H_o : \mu_1 = \mu_2$

Alternative hypothesis: $H_1 : \mu_1 \neq \mu_2$

Test statistic (C.V) : $z = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{\sigma^2}{n_1} + \dfrac{\sigma^2}{n_2}}} = \dfrac{67.5 - 68}{\sqrt{\dfrac{(2.5)^2}{1000} + \dfrac{(2.5)^2}{2000}}}$ = -5.16

$|z| = 5.16$

Tabulated value (T.V): z at 5% level of significance is 1.96

Conclusion : C.V > T.V

We reject the null hypothesis H_o.

Therefore, the samples are not drawn from the same population.

Problems for Practice:

1. A sample of 100 students is taken from a large population. The mean height of the students in this sample is 160. Can it be reasonably regarded that, in the population the mean height is 165cm and S.D is 10cm?
   Solution: Given :n = 100, $\mu = 165$, $\bar{x} = 160$, $\sigma = 10$,
   Null hypothesis : $H_O : \mu = 165$
   Alternative hypothesis: $H_1 : \mu > 165$
   Test statistic (C.V) : $z = \dfrac{\bar{x} - \mu}{\sigma / \sqrt{n}}$ = -5 ; rejected
2. In a random sample of 60 workers, the average time taken by them to get to work is 33.8minutes, with a S.D of 6.1 minutes. Can we reject the null hypothesis $\mu = 32.6$ minutes in favour of alternative hypothesis $\mu > 32.6$ minutes at tabulated value 2.58?
   Solution: Given :n = 60, $\mu = 32.6$, $\bar{x} = 33.8$, $\sigma = 6.1$,
   Null hypothesis : $H_O : \mu = 32.6$
   Alternative hypothesis: $H_1 : \mu > 32.6$
   Test statistic (C.V) : $z = \dfrac{\bar{x} - \mu}{\sigma / \sqrt{n}}$ = 1.5238 ; Accepted
3. The mean and S.D of a population are 11795 and 14054 respectively. If n=50, find 95% confidence limit for mean.
   Solution: $\bar{x} \pm 1.96 \dfrac{\sigma}{\sqrt{n}}$ = (7899.4, 15690.6)
4. An ambulance service claims that it takes on the average less than 10 minutes to reach its destination in emergency calls. A sample of 36 calls has a mean of 11 minutes and the variance of 16 minutes. Test the significance at 5% level of significance.

Solution: Given :n = 36, $\mu = 10$, $\bar{x} = 11$, $\sigma = \sqrt{16} = 4$,

Null hypothesis : $H_o : \mu = 10$

Alternative hypothesis: $H_1 : \mu \neq 10$

Test statistic (C.V) : $z = \frac{\bar{x} - \mu}{S/\sqrt{n}}$ = 1.5 ; Accepted

5. The average marks scored by 32 boys are 72 with a standard deviation of 8, while that for 36 girls is 70 with a standard deviation of 6. Test at 1% level of significance whether the boys perform better than girls.

Solution: Null Hypothesis : $H_o : \mu_1 = \mu_2$

Alternative hypothesis: $H_1 : \mu_1 \neq \mu_2$

Test statistic (C.V) : $z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}}$ = 1.15, H_0 accepted

6. An examination was given to two group of students of two colleges:
College A : mean = 75; S.D = 9; $n_1 = 50$
College B : mean = 79; S.D = 7; $n_2 = 60$
Is there a significant difference in the performance at two colleges at 5%level

Solution: Null Hypothesis : $H_o : \mu_1 = \mu_2$

Alternative hypothesis: $H_1 : \mu_1 \neq \mu_2$

Test statistic (C.V) : $z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}}$ = -2.56, H_0 rejected